

Radiation Complications and Tumor Control After Plaque Radiotherapy of Choroidal Melanoma With Macular Involvement

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• **PURPOSE:** To determine the outcome of plaque radiotherapy in the treatment of macular choroidal melanoma and to identify the risk factors associated with the development of radiation complications, tumor recurrence, and metastasis.

• **METHODS:** Chart analysis of 630 consecutive patients (630 eyes) with macular choroidal melanoma managed by plaque radiotherapy between July 1976 and June 1992.

• **RESULTS:** The median largest basal tumor diameter was 10 mm, and the median tumor thickness was 4 mm. By means of Kaplan-Meier estimates, visually significant maculopathy developed at 5 years in 40% of the patients, cataract in 32%, papillopathy in 13%, and tumor recurrence in 9%. Vision decrease by 3 or more Snellen lines was found in 40% of the patients at 5 years. Sixty-nine eyes (11%) were enucleated because of radiation complications and recurrence. Twelve percent of the patients developed metastasis by 5 years and 22% by 10 years.

Results of multivariate Cox proportional hazards analyses showed that the significant predictors for tumor recurrence were a distance of tumor margin from the optic disk of less than 2 mm ($P = .003$) and retinal invasion ($P = .009$). The significant variables that were predictive of metastasis included tumor thickness greater than 4 mm ($P = .02$) and largest basal tumor diameter greater than 10 mm ($P = .03$).

Accepted for publication Dec 31, 1998.

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Supported by TUBITAK (Dr Gündüz), the Macula Foundation, New York, New York (Drs Gündüz and C. Shields), the Paul Kayser International Award of Merit in Retina Research, Houston, Texas (Dr J. Shields), the Eye Tumor Research Foundation (Dr C. Shields), and the Pennsylvania Lions Sight Conservation and Eye Research Foundation, Philadelphia, Pennsylvania (Drs Gündüz, C. Shields, and J. Shields).

Biostatistical consultation was provided by J. Cater, PhD.

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• **CONCLUSIONS:** Plaque radiotherapy offers a 91% 5-year local tumor control rate for macular choroidal melanoma. Despite good local tumor control, the risk for metastasis is 12% at 5 years and 22% at 10 years. In 11% of the patients, enucleation eventually became necessary because of radiation complications and tumor recurrence. (Am J Ophthalmol 1999;127:579-589. © 1999 by Elsevier Science Inc. All rights reserved.)

THERE ARE SEVERAL TREATMENT METHODS FOR THE management of posterior uveal (ciliary body and choroidal) melanoma.^{1,2} Posterior uveal melanoma was treated primarily by enucleation several years ago, but concern about possible acceleration of metastasis after enucleation³ brought about a trend toward instituting radiotherapy. A number of studies in which patients were not randomly assigned have shown similar survival of patients treated with numerous methods, including enucleation, plaque radiotherapy, and charged particle radiotherapy.⁴⁻⁶ The Collaborative Ocular Melanoma Study is currently evaluating in a randomized fashion the effect of enucleation versus plaque radiotherapy in patients with choroidal melanoma.⁷

Plaque radiotherapy has become an established method of treatment for posterior uveal melanoma. One of the concerns regarding plaque radiotherapy is that it may be unsuitable for the treatment of choroidal melanoma with macular involvement, because of the resultant severe visual loss.⁸ We review our experience with plaque radiotherapy to treat choroidal melanoma involving the macula with respect to risk factors for the development of radiation events, including complications and visual decrement and tumor events of recurrence and metastasis.

PATIENTS AND METHODS

WE ANALYZED THE RECORDS OF ALL PATIENTS WHO HAD posterior uveal melanoma with macular involvement treated with plaque radiotherapy on the Ocular Oncology